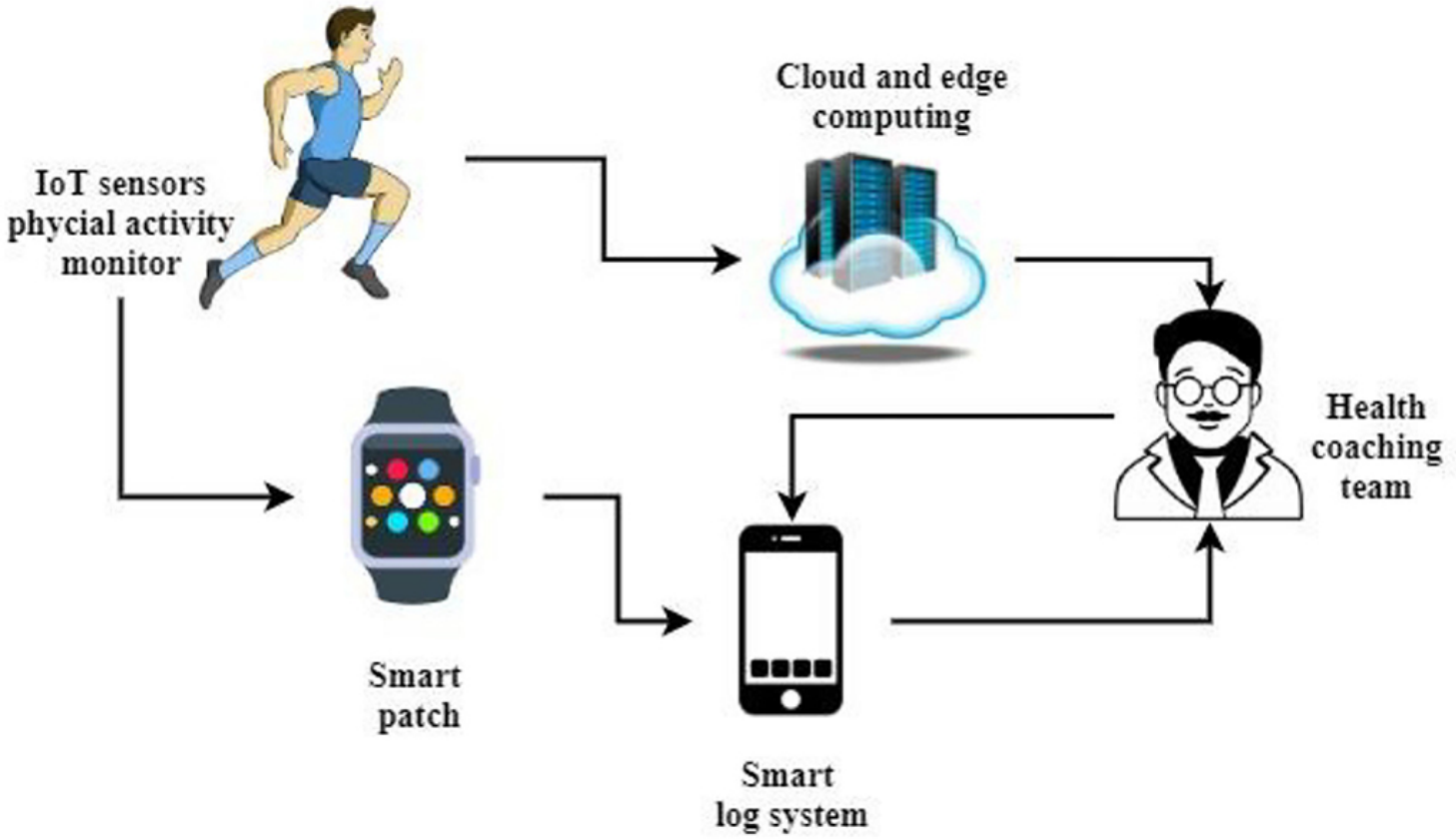
HEALTH AND WELL-BEING: HEALTH AND FITNESS TRACKING DEVICES AND APPLICATIONS

Google Fit, Samsung Health, Apple Health, Xiaomi, Mi Band, Run Keeper

INTRODUCTION TO ICT IN HEALTH AND WELL-BEING

- **ICT** (Information and Communication Technology) plays a transformative role in healthcare, facilitating new approaches to health, wellness, and fitness.
- The use of ICT in health has grown due to increased digitalization, the need for accessible health solutions, and the rise of wearable technologies.
- ICT applications in health focus on **prevention, early detection, and management** of health conditions by tracking vital signs, physical activity, and overall health metrics.

ICT IN HEALTH AND WELL-BEING



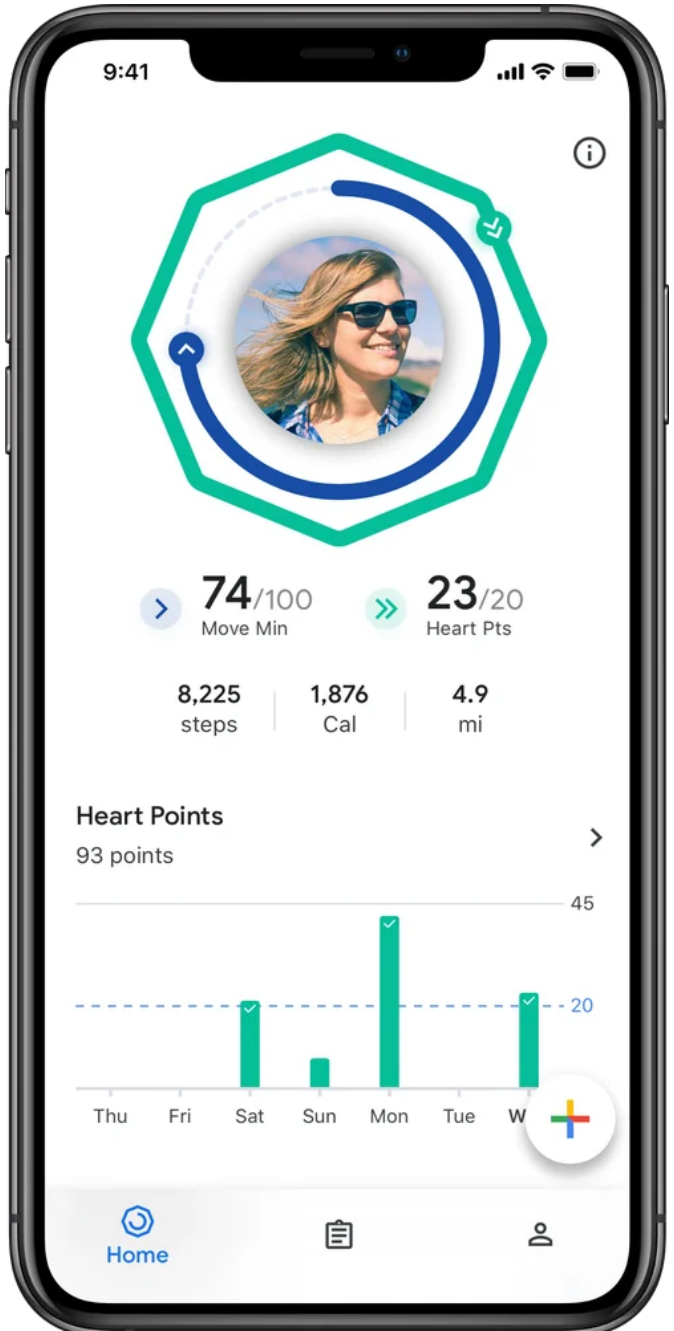
OVERVIEW OF HEALTH AND FITNESS TRACKING DEVICES AND APPS

- Fitness tracking devices and applications have gained popularity as tools for monitoring and improving personal health.
- These devices use sensors to capture data like heart rate, sleep patterns, step count, and more.
- They integrate with **smartphone applications** for deeper analysis, tracking history, and goal setting, providing users with insights into their health and fitness progress.

POPULAR HEALTH AND FITNESS TRACKING PLATFORMS

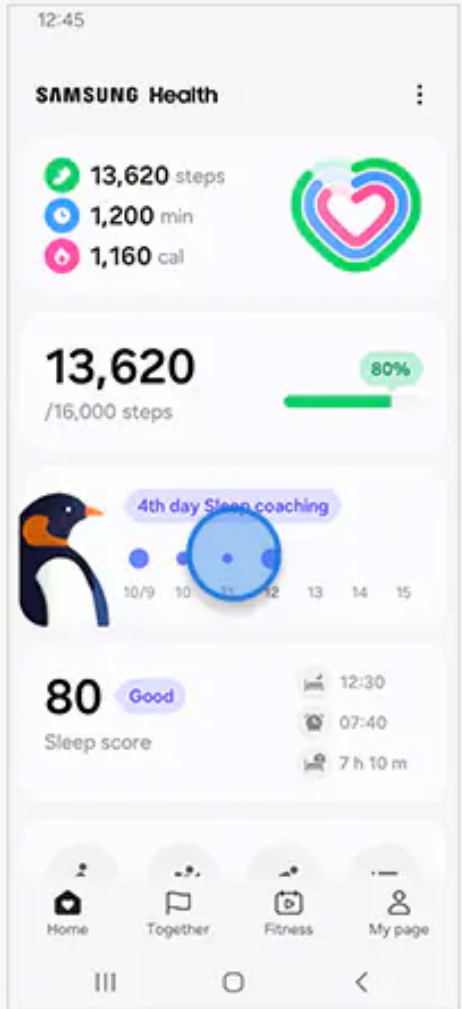
Google Fit

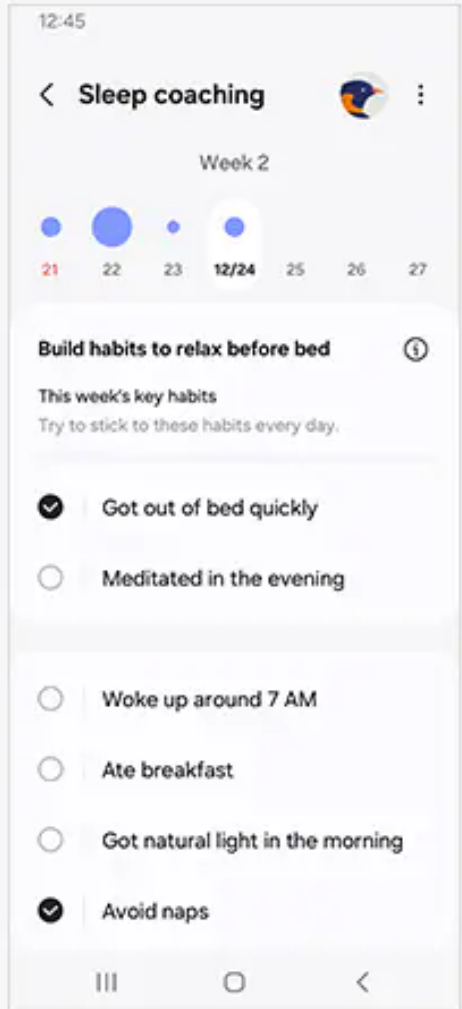
- Developed by **Google**, integrates with Android
- Tracks physical activities, such as walking, running, and cycling, using smartphone and wearable sensors.
- Offers personalized goals and “Heart Points” to encourage users to meet WHO activity recommendations.
- Integrates with third-party apps like **Strava** and **MyFitnessPal**.

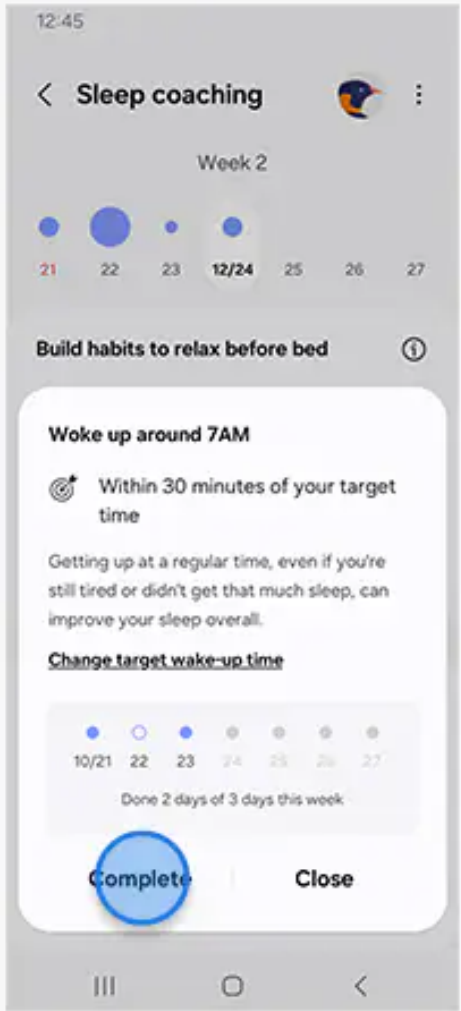

9:41
74/100
Move Min
23/20
Heart Pts
8,225
steps
1,876
Cal
4.9
mi
Heart Points
93 points
45
20
Thu
Fri
Sat
Sun
Mon
Tue
W
Home

SAMSUNG HEALTH

- Developed by **Samsung** and compatible with Galaxy devices and other Android smartphones.
- Tracks a wide range of metrics: steps, heart rate, stress levels, sleep, and nutrition.
- Provides guided workouts, community challenges, and fitness programs.
- Compatible with Samsung Galaxy Watches and many other devices.


12:45
SAMSUNG Health
13,620 steps
1,200 min
1,160 cal
13,620
/16,000 steps
80%
4th day Sleep coaching
10/9 10 11 12 13 14 15
80
Good
Sleep score
12:30
07:40
7 h 10 m
Home
Together
Fitness
My page


12:45
Sleep coaching
Week 2
21 22 23 12/24 25 26 27
Build habits to relax before bed
This week's key habits
Try to stick to these habits every day.
Got out of bed quickly
Meditated in the evening
Woke up around 7 AM
Ate breakfast
Got natural light in the morning
Avoid naps


12:45
Sleep coaching
Week 2
21 22 23 12/24 25 26 27
Build habits to relax before bed
Woke up around 7AM
Within 30 minutes of your target time
Getting up at a regular time, even if you're still tired or didn't get that much sleep, can improve your sleep overall.
Change target wake-up time
10/21 22 23
Done 2 days of 3 days this week
Complete
Close

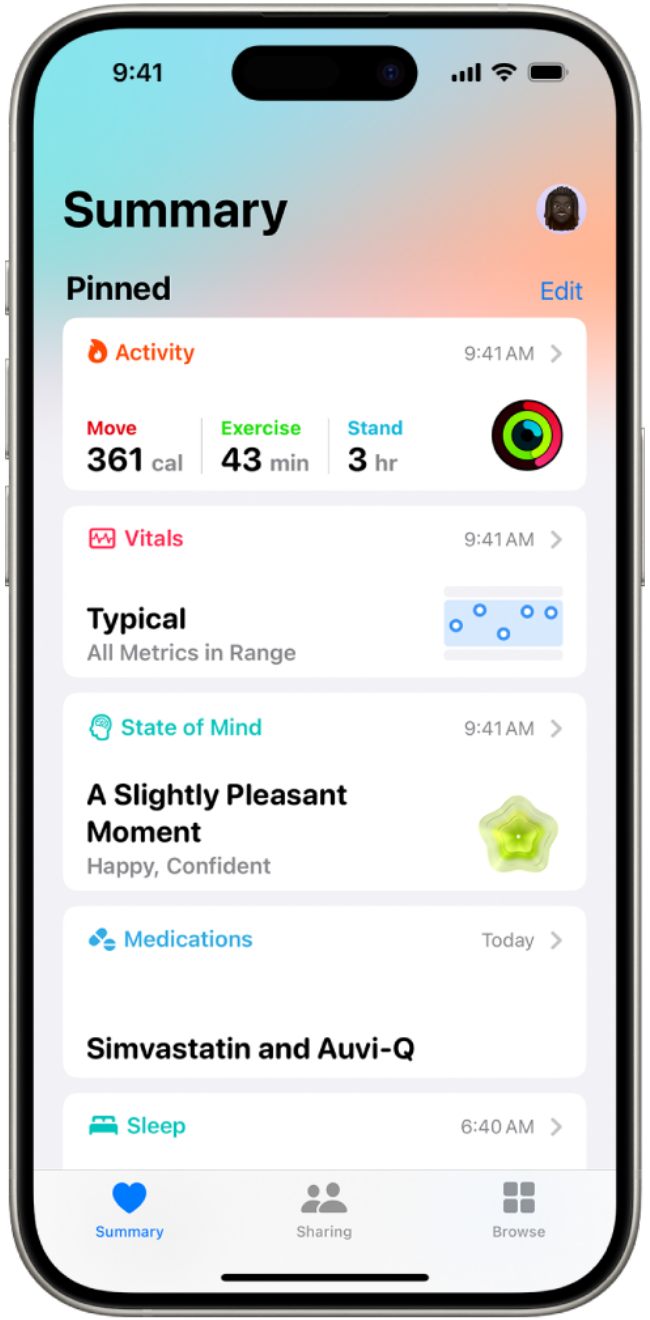
APPLE HEALTH

Built into Apple devices, specifically iPhone and Apple Watch.

Offers a comprehensive dashboard for tracking fitness, activity, and various health metrics.

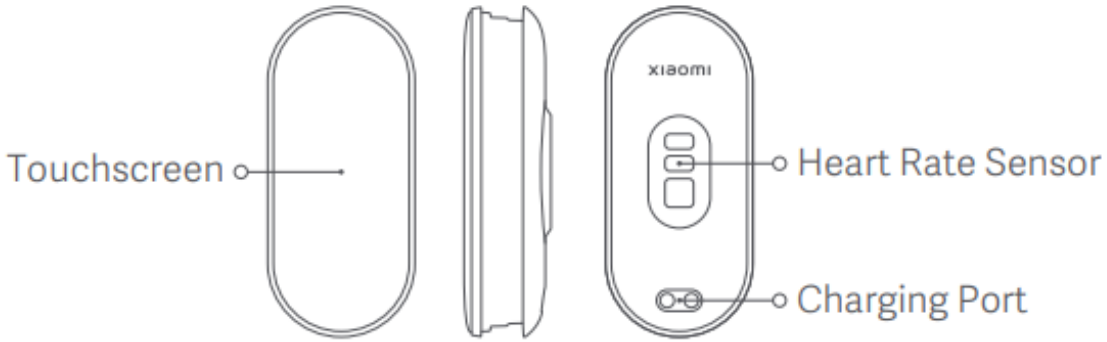
Integrates with a wide range of third-party health apps and Apple Watch sensors, including ECG monitoring

Offers “Trends” insights for long-term health analysis and an emergency health profile.


9:41
Summary
Pinned
Edit
Activity
9:41 AM
Move
361 cal
Exercise
43 min
Stand
3 hr
Vitals
9:41 AM
Typical
All Metrics in Range
State of Mind
9:41 AM
A Slightly Pleasant Moment
Happy, Confident
Medications
Today
Simvastatin and Auvi-Q
Sleep
6:40 AM
Summary
Sharing
Browse

XIAOMI MI BAND

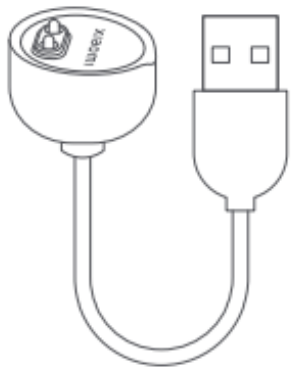
- Affordable fitness tracker with a large user base, especially in Asia.
- Tracks steps, sleep quality, heart rate, and workouts, and syncs with the **Mi Fit** app.
- Provides long battery life and budget-friendly pricing, making it accessible to many users.
- Also works with Google Fit for users wanting to merge data.



ⓐ Fitness Tracker



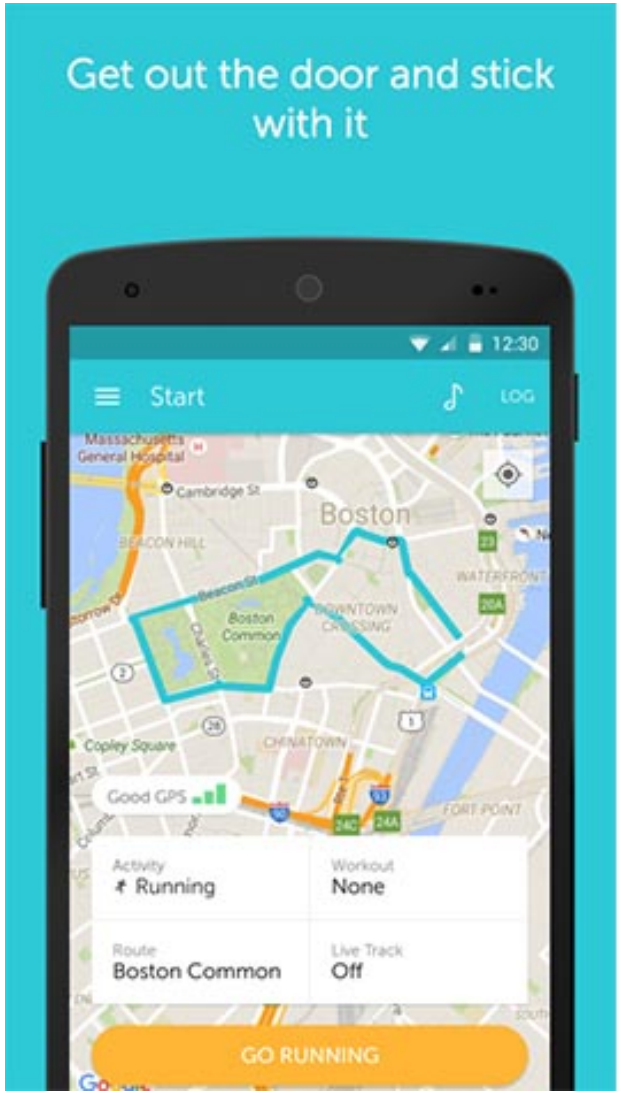
ⓑ Wristband



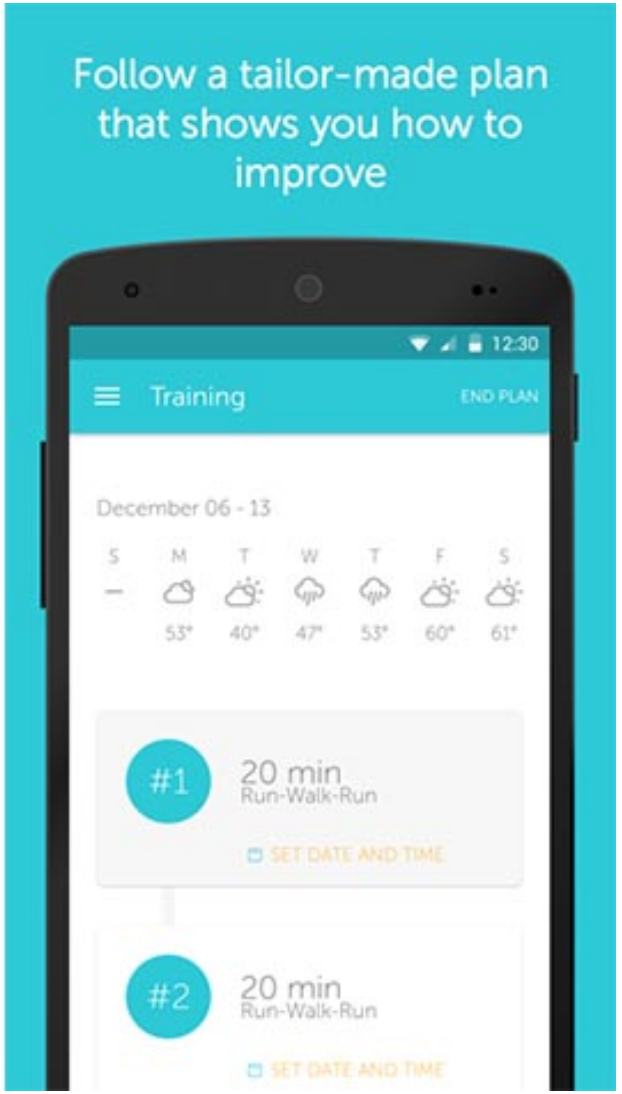
ⓒ Charging Cable

RUN KEEPER

- Popular app focused on **running and outdoor activities**; uses GPS to track routes and progress.
- Tracks distance, pace, and calories burned, and offers personalized training plans.
- Integrates with other health apps and devices to expand its functionality beyond running.


Get out the door and stick with it
12:30
Start
LOG
Boston
Good GPS
Activity
Running
Workout
None
Route
Boston Common
Live Track
Off
GO RUNNING


Track your workouts in real time and get audio updates along the way
12:30
Time
22:21
Good GPS
9:56
Avg min/mi
246
Calories
Distance
2.21
miles
Current Pace
9:46
min/mi
STOP
PAUSE


Follow a tailor-made plan that shows you how to improve
12:30
Training
END PLAN
December 06 - 13
S M T W T F S
53° 40° 47° 53° 60° 61°
#1
20 min
Run-Walk-Run
SET DATE AND TIME
#2
20 min
Run-Walk-Run
SET DATE AND TIME

FEATURES OF HEALTH AND FITNESS TRACKING APPLICATIONS

Activity Tracking: Monitoring steps, distance, calories burned, and physical activities like running, cycling, and walking.

Heart Rate Monitoring: Sensors on wearable allow continuous or on-demand heart rate tracking, crucial for managing stress and cardiovascular health.

Sleep Analysis: Trackers monitor sleep patterns and stages, providing insights for better sleep quality.

Calorie Counting and Nutrition Tracking: Many apps allow users to log food intake and calories, providing nutrition guidance for a balanced diet.

Goal Setting and Feedback: Setting fitness goals and receiving feedback on progress is a key motivational feature.

Health Metrics Visualization: Visual representations of data (charts, trends) help users easily interpret their health over time.

IMPACT OF HEALTH TRACKING DEVICES ON WELL-BEING

Increased Awareness: These devices make users more aware of their health behaviors, helping them make informed lifestyle choices.

Encouragement of Healthy Habits: Through reminders, goal-setting, and community challenges, users are encouraged to stay active.

Remote Health Monitoring: Some devices can share data with healthcare providers, facilitating remote monitoring for chronic conditions.

Personalized Health Insights: Data analysis in these apps allows personalized recommendations, helping users optimize their health routines.

PRIVACY AND DATA SECURITY CONCERNS

Data Collection and Storage: These applications collect a vast amount of personal data, raising privacy concerns.

Third-party Data Sharing: Users need to be mindful of app permissions and data-sharing practices with third parties.

Security Measures: Many companies implement data encryption, two-factor authentication, and user-controlled data access.

THE FUTURE OF HEALTH AND FITNESS TRACKING

AI and Machine Learning Integration: Future devices will offer more sophisticated insights with AI analyzing trends and providing predictive health advice.

Wearable Advancements: Wearable's will likely evolve to provide non-invasive glucose monitoring, blood pressure tracking, and more.

Enhanced Remote Healthcare Support: Health trackers may further integrate with telemedicine, enabling real-time health monitoring and consultations.

Focus on Mental Health: Apps may include more mental health features, like stress management tools and mindfulness exercises.